

Digital Circuits and VHDL – Overview and Examples

Florian Hagemann

Abstract—Here be an abstract...

CONTENTS

I	Introduction	1
II	Digital Circuits	1
II-A	Electronic Switches: Transistors	1
II-B	Making Computers Think	1
II-C	Examples	1
III	Introduction to VHDL	1
III-A	Hardware Description Languages	1
III-B	Basic Syntax	1
III-C	Blackbox Diagrams and Entities	1
IV	Combinational Circuits	1
IV-A	Basic Logic Gates	1
V	Sequential Circuits	1
VI	Finite State Machines (FSM)	1
VI-A	Moore Machines	1
VI-B	Mealy Machines	1
	References	1

I. INTRODUCTION

II. DIGITAL CIRCUITS

WHAT is a *digital circuit*? To understand this, we need to examine each word separately. A *circuit* is a closed loop that allows electricity to flow. Typically it consists of a power source and some device (e.g., a motor, LED, etc.) that uses that electricity. A circuit needs to be closed in order for the electricity to flow through it [1]. *Digital* comes from the Latin “*digitalis*” meaning finger or toe. Digital can mean “something to do with digits (fingers or toes)” but can also mean the “property of representing values as discrete, often binary, numbers rather than a continuous spectrum” [2].

A **digital circuit** is a closed loop that electricity can flow through that uses discrete values. Most often these values are ‘1’ (on, high, or true) and ‘0’ (off, low, or false)¹. This system of using two values is called *binary*.

We can now look at a very basic example (Fig. 1) of what we can do by defining binary values. When the switch is

closed, electricity can flow through the LED and it will light up; if it is not closed, no electricity will flow.

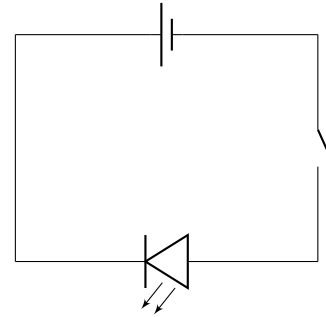


Fig. 1. Simple series circuit with a voltage source, switch, and LED.

We only care for the switch and the output state of the circuit though, not what the power source is or what the output is connected to, so we can simplify our diagram (Fig. 2).

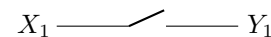


Fig. 2. Simplified diagram of Fig. 1.

A. *Electronic Switches: Transistors*

B. *Making Computers Think*

C. *Examples*

III. INTRODUCTION TO VHDL

A. *Hardware Description Languages*

B. *Basic Syntax*

C. *Blackbox Diagrams and Entities*

IV. COMBINATIONAL CIRCUITS

A. *Basic Logic Gates*

V. SEQUENTIAL CIRCUITS

VI. FINITE STATE MACHINES (FSM)

A. *Moore Machines*

B. *Mealy Machines*

REFERENCES

- [1] Wiktionary contributors. circuit. Wiktionary, the free dictionary. [Online]. Available: <https://en.wiktionary.org/wiki/circuit>
- [2] —. digital. Wiktionary, the free dictionary. [Online]. Available: <https://en.wiktionary.org/wiki/digital>
- [3] B. Mealy and F. Tappero, *Free Range VHDL*. Free Range Factory, 2025.
- [4] D. M. Harris and S. L. Harris, *Digital Design and Computer Architecture: RISC-V Edition*, 2nd ed. Elsevier, 2022.
- [5] M. M. Mano and M. D. ciletti, *Digital Design*, 5th ed. Pearson, 2013.

Thanks to ...

¹You can use more than two values but binary is most common. See https://en.wikipedia.org/wiki/Ternary_computer